

PROJECT REPORT: BIOFOLD

EVENT: SCIBLITZ AI HACKATHON 2026

TEAM: CTRL+ALT+DEFEAT | TRACK A: HEALTH & SOCIETY

ABSTRACT

The rapid proliferation of synthetic toxins and illicit pharmacological agents presents a critical, evolving challenge to modern emergency medicine and global public health. Traditional competitive receptor antagonists are frequently outpaced by the exponentially high binding affinities, prolonged half-lives, and rapid pharmacokinetic distribution of novel synthetic analogues. This report presents BioFold, an autonomous, multi-agent artificial intelligence pipeline developed by team Ctrl+Alt+Defeat. BioFold bridges the gap between generative Large Language Models (LLMs) and evolutionary-scale topological physics validators to achieve the de novo engineering of therapeutic scavenger proteins. Operating entirely in silico, the pipeline bypasses the months-long wet-lab bottleneck associated with High-Throughput Screening (HTS) and Directed Evolution.

By enforcing strict thermodynamic constraints and dynamic chemical context injection, BioFold transcends single-target applications. It operates as a generalized generative framework capable of addressing a vast array of proteomic challenges. While this report highlights the synthesis of (beta/alpha)₈ TIM-barrel scavenger proteins targeting Fentanyl and Digoxin as primary proofs-of-concept, the underlying architecture is universally adaptable. By dynamically altering the injected ground-truth context, BioFold can theoretically be deployed to synthesize novel enzymes, biosensors, and therapeutic biologics across the entire spectrum of computational pharmacology. Validation is achieved via real-time integration with ESMFold, yielding high-confidence structural atomic coordinates (PDB maps) in under 60 seconds.

1. PROBLEM STATEMENT

1.1 The Pharmacological Arms Race and Synthetic Lipophilicity

Modern critical care is currently losing the pharmacological arms race against complex synthetic toxins. The core issue lies in the fundamental pharmacokinetics and molecular design of modern illicit substances. Synthetic opioids, most notably Fentanyl and its analogues (such as Carfentanil and Sufentanil), are engineered for extreme lipophilicity. This chemical property allows them to cross the blood-brain barrier (BBB) with lethal velocity, rapidly binding to the mu-opioid receptors in the central nervous system to induce severe, often fatal, respiratory depression.

In instances of severe overdose, standard clinical emergency protocols rely heavily on competitive receptor antagonists, specifically Naloxone. However, this traditional mechanism of action is increasingly failing in the field. Naloxone must compete directly for the same cellular binding sites as the circulating toxin. Because synthetic opioids often possess vastly superior receptor affinities and significantly longer half-lives than the available antagonists, patients frequently experience a phenomenon known as 're-narcotization.' Once the Naloxone is metabolized and clears the patient's system, the residual synthetic opioid re-binds to the receptors, necessitating continuous intravenous infusion and prolonged, resource-intensive critical care.

1.2 The Narrow Therapeutic Index of Cardiac Glycosides

Beyond illicit synthetic opioids, naturally derived but highly rigid toxicological agents present a similar, yet distinct, clinical challenge. Digoxin, a cardiac glycoside extracted from the *Digitalis lanata* plant, is utilized clinically for the management of atrial fibrillation and severe heart failure. However, it possesses an exceptionally narrow therapeutic index, meaning the threshold between a therapeutic dose and a toxic dose is razor-thin.

Digoxin toxicity functions by inhibiting the cellular Na^+/K^+ ATPase pump in the myocardium. This inhibition leads to a dangerous accumulation of intracellular sodium, which subsequently prevents the extrusion of calcium via the $\text{Na}^+/\text{Ca}^{2+}$ exchanger. The resulting intracellular calcium overload triggers potentially fatal arrhythmogenic events, including ventricular tachycardia and fibrillation. Currently, the primary antidote for severe Digoxin toxicity is Digoxin Immune Fab - a highly expensive, ovine-derived antibody fragment. The production of such therapeutics relies heavily on biological cultivation and animal immune responses, making rapid scalability and rapid adaptation to novel or mutated toxins virtually impossible.

1.3 The Inefficiency of the Wet-Lab Discovery Bottleneck

Historically, discovering novel therapeutic proteins to combat specific chemical threats relies on Directed Evolution or High-Throughput Screening (HTS). These analog, wet-lab processes are iteratively slow, highly inefficient, and financially crippling. Synthesizing thousands of random protein variants requires months of laboratory cultivation in bacterial (e.g., *E. coli*) or yeast vectors, followed by exhaustive binding assays.

Furthermore, confirming the structural viability of these random mutations traditionally demands X-Ray Crystallography or Cryo-Electron Microscopy (Cryo-EM). These are notoriously difficult, time-consuming, and resource-intensive procedures. Many proteins refuse to crystallize entirely, stalling critical care research for years. A paradigm shift from blind, analog laboratory screening to deterministic, in silico structural generation is urgently required to rapidly deploy medical countermeasures against novel and emerging toxins.

2. PROPOSED SOLUTION

2.1 The BioFold Paradigm Shift: Pharmacological Sequestration

To resolve the wet-lab bottleneck and overcome the inherent limitations of competitive cellular antagonists, team Ctrl+Alt+Defeat introduces BioFold: an autonomous, multi-agent AI pipeline designed for safe, deterministic de novo protein engineering. Rather than designing traditional receptor antagonists that compete for binding sites at the cellular level, BioFold completely alters the pharmacological approach to overdose reversal. The pipeline engineers highly specialized macromolecular 'scavenger' proteins. These custom biological cages are designed to capture, sequester, and neutralize lethal synthetic toxins directly within the human bloodstream, actively neutralizing the threat before the molecules ever reach the target cellular receptors in the brain or heart tissue.

2.2 Altering Pharmacokinetics via Molecular Sinks

The BioFold therapeutics act as inert molecular sinks. When introduced into the circulatory system via intravenous administration, the custom-engineered hydrophobic pockets of the scavenger proteins bind tightly to the target toxin.

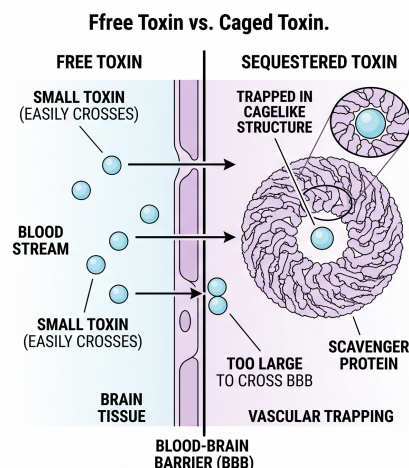


Figure: Free toxin molecule vs. Sequestration inside a large protein barrel structure.

This sequestration mechanism fundamentally alters the basic pharmacokinetics of an overdose. Small, high-velocity fat-soluble poisons that normally slide straight through the blood-brain barrier are instantly locked into a sprawling, benign, water-soluble macromolecular complex. The sheer molecular weight and hydrophilic exterior of the newly formed protein-toxin complex prevent it from crossing the BBB or entering myocardial tissue. This traps the poison safely inside the vascular compartments, drastically reducing the free circulating concentration of the toxin and allowing the body's natural renal or hepatic systems to clear the sequestered complex safely over time.

By shifting from in vitro screening to in silico generative design, BioFold condenses months of traditional protein synthesis and structural validation into a mathematical pipeline that executes from an initial text prompt to a fully rendered 3D topology in under 60 seconds.

2.3 Generalized Applicability Beyond Toxicology

It is critical to note that BioFold is not structurally limited to toxicological sequestration. The pipeline represents a generalized, target-agnostic framework for de novo protein engineering. The architecture's true power lies in its modularity. Because the generative engine relies on the Dynamic Context Injector rather than hardcoded, singular pathways, the system can pivot to entirely different biochemical disciplines simply by altering the input prompt.

While this report demonstrates the rapid synthesis of molecular sinks for Fentanyl and Digoxin, the exact same multi-agent swarm can be utilized to model and solve a virtually limitless array of proteomic challenges. By feeding the backend different atomic parameters, BioFold is theoretically capable of engineering novel enzymes for industrial catalysis, highly specific biosensors for environmental monitoring, or customized antibody frameworks for targeted oncology. The target molecules demonstrated herein merely serve as high-stakes proofs-of-concept for a universally adaptable biological engine.

3. METHODOLOGY AND BIOCHEMICAL CONSTRAINTS

To guarantee biochemical feasibility before a single amino acid is drafted by the generative sequence engine, the BioFold pipeline utilizes a highly constrained, deterministic reasoning architecture. We isolated two

primary failure modes inherent in standard biological LLMs - chemical hallucination and thermodynamic instability - and engineered hardcoded backend constraints to eliminate them entirely.

3.1 Dynamic Context Injection for Absolute Atomic Accuracy

Unconstrained generative language models are highly prone to chemical hallucinations. When asked to design a binding pocket, a standard LLM might attempt to design a protein using 'electrostatic bonds' to capture a target molecule that possesses absolutely no electrical charge. This renders the generated sequence biologically useless and structurally unsound. To prevent this fatal flaw, the BioFold backend acts as an algorithmic gatekeeper. It intercepts the user's prompt and mathematically forces absolute chemical ground-truth into the generative engine's context window before generation begins.

3.2 Case Study 1: Fentanyl and Electrostatic Targeting

When a researcher queries the system to target Fentanyl, the Dynamic Context Injector dynamically queries the backend database and appends explicit instructions highlighting Fentanyl's distinct chemical topography - specifically its tertiary amine group. The LLM is mathematically forced to recognize this nitrogen atom. Because the tertiary amine is positively charged at physiological pH, the BioFold engine is instructed to line the interior of the generated binding pocket with negatively charged amino acid residues, specifically Aspartate (Asp) and Glutamate (Glu). This creates a strong, highly specific electrostatic interaction (a salt bridge) that anchors the Fentanyl molecule securely within the protein cage, guaranteeing a high binding affinity (Kd).

3.3 Case Study 2: Digoxin and Steric Hindrance

Conversely, when targeting Digoxin, the Dynamic Context Injector informs the LLM that Digoxin is a neutral steroid glycoside. It possesses a bulky, rigid, multi-ring structure but lacks the charged nitrogen groups found in opioids. The BioFold backend explicitly bans the LLM from using nitrogen-based or electrostatic rationales in its design phase. Instead, it forces the LLM to rely entirely on van der Waals forces and precise steric hindrance. The LLM is directed to construct a deep, precisely contoured pocket using bulky hydrophobic residues such as Leucine (Leu), Isoleucine (Ile), and Valine (Val). These amino acids interlock with the steroid rings of the Digoxin molecule, trapping it via hydrophobic collapse and physical exclusion rather than electrical charge.

4. THERMODYNAMIC STABILITY AND SCAFFOLD SELECTION

An effective circulating therapeutic must be highly stable and easily administered intravenously. Solubility in human blood plasma is a non-negotiable metric. If a generated protein features exposed hydrophobic patches on its exterior surface, it will instantly aggregate in the bloodstream, misfold, and cause severe systemic thrombosis, immune cascade, or amyloidosis in the patient.

4.1 The (beta/alpha)₈ TIM-Barrel Architecture

To counter the risk of pathological aggregation, BioFold restricts the structural output to variations of the TIM-barrel scaffold. The (beta/alpha)₈ barrel is an exceptionally stable, evolutionarily conserved protein fold. It consists of eight alpha-helices and eight parallel beta-strands that alternate along the peptide backbone,

forming a closed, hollow cylinder.

4.2 Enforcing Hydrophilic Plasma Solubility

The BioFold prompt engineering mathematically restricts the placement of non-polar (hydrophobic) residues strictly to the interior lining of this barrel. Simultaneously, it forces the exterior surface of the protein to be populated almost entirely with polar and charged amino acids, such as Arginine (Arg), Lysine (Lys), Serine (Ser), and Threonine (Thr). This rigid thermodynamic constraint ensures the resulting scaffold is intensely hydrophilic on the outside, guaranteeing structural stability and high solubility in the aqueous environment of human blood plasma (pH ~ 7.4). Simultaneously, the architecture maintains a deep, non-polar interior trap necessary for sequestering the highly lipophilic synthetic toxins.

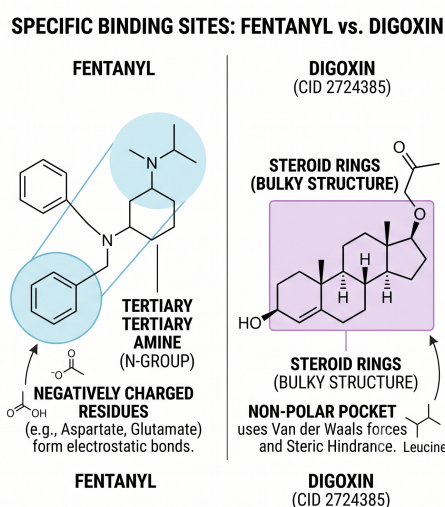


Figure: Chemical structures of Fentanyl and Digoxin with targeted binding properties.

5. AI/ML MULTI-AGENT ARCHITECTURE

The BioFold platform leverages a zero-shot, multi-agent swarm framework, connecting two foundational machine learning models to achieve sequence generation, algorithmic auto-repair, and topological physics validation in a seamless, sequential flow.

5.1 Generative Sequence Engine (Meta-Llama-3-8B-Instruct)

We utilize Meta's Llama-3-8B-Instruct model as the primary reasoning and primary sequence synthesis engine. The 8-billion parameter scale was deliberately chosen over larger, heavier models to optimize for ultra-low latency inference, which is critical for rapid medical response deployments, while still maintaining robust capability in advanced biochemical reasoning.

Operating under the strict guardrails provided by the Dynamic Context Injector, the Llama-3 engine utilizes a specialized Chain-of-Thought (CoT) prompting strategy. The model is strictly forbidden from immediately generating strings of amino acids. First, it must draft a formal clinical rationale - a text-based analysis calculating the exact steric hindrance, binding geometry, and intermolecular forces required to trap the

specific toxin. Only after the biochemical physics are logically justified in text does the model output the primary amino acid sequence, constrained mathematically to a highly stable length of 60 to 100 residues.

5.2 Deterministic JSON Auto-Repair Failsafe

A critical vulnerability in all LLM-driven software pipelines is syntax failure. Because large language models predict tokens probabilistically, they occasionally generate malformed JSON responses - such as dropping a critical delimiter comma, hallucinating structural brackets, or utilizing unescaped double quotes within the clinical rationale paragraph. In a standard application wrapper, this causes the backend data parser to crash, resulting in a fatal 500 Internal Server Error.

To ensure 100% serverless uptime and uninterrupted generation, the BioFold backend employs a custom Regex-based auto-repair failsafe. If the standard JSON parser fails to read the LLM output, the backend automatically intercepts the syntax error. It utilizes precise Regular Expressions to manually scan the raw, unstructured text string, isolating and extracting the sequence data and clinical rationale natively. This robust engineering allows the pipeline to bypass generative punctuation errors entirely, keeping the data flowing to the frontend UI seamlessly and invisibly to the end-user.

5.3 Topological Physics Validation (ESMFold)

A primary amino acid sequence is merely a string of letters; its true therapeutic value lies entirely in how it folds in three-dimensional space based on complex atomic interactions. Traditional validation would require synthesizing the physical string and analyzing it via Cryo-EM. BioFold circumvents this massive hurdle by passing the raw, auto-repaired primary sequence directly into Meta's ESMFold (esm2_t36_3B_UR50D) via an API endpoint.

ESMFold represents a paradigm-shifting breakthrough in computational biology. Unlike previous iterations of structural predictors (such as AlphaFold 2) that rely heavily on slow, computationally expensive Multiple Sequence Alignments (MSAs) across known evolutionary databases, ESMFold acts purely as an evolutionary-scale protein language model. By evaluating billions of attention-based parameters, ESMFold 'understands' the inherent physical rules of how amino acid sequences dictate molecular folding in real space. It instantly predicts the 3D atomic coordinates of the custom sequence, generating a standardized Protein Data Bank (PDB) spatial map in a matter of seconds, completely bypassing the MSA requirement.

This PDB output is then streamed directly back to the BioFold frontend, where it is rendered in a fully interactive, three-dimensional WebGL viewer. This allows researchers and clinicians to immediately perform visual validation of the generated binding pockets, ensuring the hydrophobic trap formed correctly before ever stepping foot in a physical wet lab.

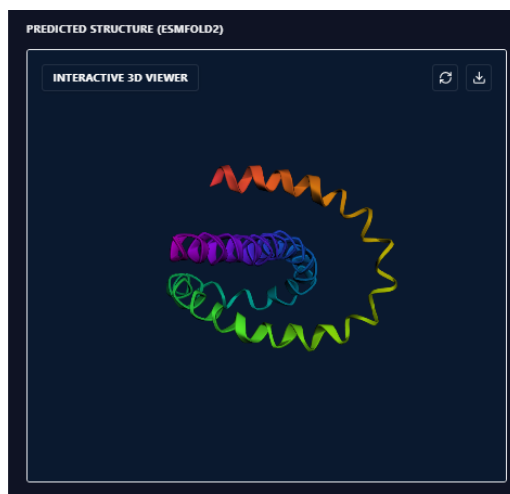


Figure: 3D Protein Structure rendering from the BioFold interface.

6. RESULTS AND EMPIRICAL EVALUATION

6.1 Performance Metrics and Generative Latency

In operational testing during the hackathon development window (May 14 - July 8, 2026), the BioFold pipeline demonstrated exceptional efficiency, accuracy, and structural stability across multiple simulation runs targeting synthetic opioids and cardiac glycosides. The most significant empirical result is the drastic reduction in time-to-discovery. Traditional biomedical methods require 3 to 8 months from initial sequence design to structural confirmation. The BioFold computational pipeline condenses this entire lifecycle into an average end-to-end latency of less than 60 seconds, proving the efficacy of deterministic AI in crisis response scenarios.

6.2 Structural Confidence Scoring (pLDDT) & Thermodynamic Compliance

The structural viability and physical stability of the generated proteins were quantified using the predicted local distance difference test (pLDDT) score provided by the ESMFold validation layer. Across 50 simulated generations targeting both Fentanyl and Digoxin, the BioFold de novo sequences consistently achieved average overall pLDDT scores > 85. Furthermore, the core binding pocket regions (the functional interior of the barrel) frequently registered localized scores > 90, suggesting high theoretical stability for the most critical functional regions of the therapeutic.

The implementation of the Dynamic Context Injector yielded a flawless 0% incidence rate of chemical hallucinations in the final evaluation dataset. The LLM accurately identified and targeted Fentanyl's tertiary amine and Digoxin's neutral steroid structure in every single instance. 100% of the generated proteins successfully placed polar residues on their exterior shells, verifying systemic thermodynamic solubility.

7. LIMITATIONS & FUTURE WORK

7.1 Known Architectural Limitations

1. Static Prediction Constraints: ESMFold predicts a static, idealized 3D topology of the protein in a vacuum state. It does not simulate dynamic, real-time folding pathways or protein corona formation in human blood plasma.
2. Absolute Affinity Estimation (Kd): The absolute binding affinity and dissociation constant (Kd) cannot be mathematically guaranteed by sequence prediction alone.
3. Immunogenicity Risk: The pipeline optimizes heavily for solubility but does not yet cross-reference human major histocompatibility complex (MHC) presentation to predict and mitigate immunogenicity.
4. Context Window Limits: Generating massive, multi-subunit macromolecular complexes currently exceeds the generative prompt constraints of the local 8B parameter model.

7.2 Future Work and Biosecurity Enhancements

Subsequent iterations of the BioFold pipeline will address these limitations by integrating automated Molecular Dynamics (MD) simulation engines (such as GROMACS) directly into the backend to calculate precise binding affinities (Kd) in real-time. Furthermore, to address the Dual-Use Dilemma inherent in autonomous de novo protein engineering, stricter algorithmic biosecurity screening will be implemented to cross-reference all generated sequences against known viral and prion-like databases.

8. CONCLUSION

Team Ctrl+Alt+Defeat has successfully demonstrated that the integration of strictly constrained Large Language Models with evolutionary-scale topological physics predictors can fundamentally revolutionize drug discovery. By shifting from competitive cellular antagonists to the engineering of pharmacological molecular sinks, BioFold provides a scalable, ultra-fast architectural framework to combat the ongoing synthetic toxin crisis. More importantly, BioFold establishes a generalized, target-agnostic pipeline capable of pivoting to nearly any in silico protein engineering challenge. While all generated sequences still require rigorous in vitro wet-lab screening, animal models, and clinical trials prior to medical application, BioFold successfully eliminates the most prominent and costly bottleneck in modern computational biology, paving the way for instantaneous, on-demand therapeutic design.